



Experimental study of thermal damage to in vitro skin tissue welding by femtosecond laser



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Highlights

- Quantifying thermal damage during femtosecond laser tissue welding.
- Find the correspondence between peak temperature and microstructure texture characteristic values.
- Best parameter combination found for femtosecond laser skin welding.

Abstract

This study aims to investigate and evaluate the thermal damage changes during femtosecond laser welding in vitro skin tissue. Based on the orthogonal test, a grey-scale co-occurrence matrix was used to extract the textural characteristic values of the microscopic tissue and to find the correspondence between the peak temperature and the textural characteristic values of the in vitro skin tissue. The statistical analysis software Minitab was used to design an orthogonal test, with laser power, laser scanning speed, off-focus amount, and laser scanning times as independent variables and the degree of thermal damage of in vitro skin tissue as the dependent variable, to establish the association between the four influencing factors and the degree of thermal damage. The microstructure of the laser-welded in vitro skin tissue is close to normal tissue with little thermal damage and high tensile strength when the laser power is 14W, the scanning speed is 50mm/s, and the off-defocus amount is -1mm, and the laser scanning times are 75. Extreme difference analysis was used to identify significant parameters impacting heat damage in skin tissue, and the associated mechanisms were analyzed. This research can be utilized to better understand and improve the quality of ultrashort pulse laser welding to biological tissues.



Keywords

Femtosecond laser; Thermal damage; Textural characteristic value

1. Introduction

Laser tissue welding (LTW) is an emerging surgical wound-healing technology that can replace traditional surgical wound-healing techniques like sutures, anastomotic staples, and biological clips. LTW is performed by thermal changes in the molecular structure of the tissue caused by the energy of the laser acting on the biological tissue. The application of laser in the connection of blood vessels and organs has been studied extensively [1], [2], [3], [4], [5]. Water, collagen, and elastin are primarily involved in the welding of these tissues. Water molecules in living tissues partially dissolve during LTW by absorbing laser energy and

heating the collagen helix. As the biological tissue cools, the protein molecules within the tissue recombine [6], [7]. Laser-welded biological tissues have advantages over traditional means of surgical wound healing in terms of shorter operative time, less inflammatory response, no rejection, shallow postoperative scarring, and immediate intraoperative watertight closure to reduce the formation of scar tissue [8], [9], [10]. A laser is used to scan the edges of the tissue incision during the LTW technique. The depth of light penetration is determined by the scattering length and absorption coefficient of the tissue, allowing the thickness of the laser-connected biologic tissue to be reached. When the laser's penetration depth matches the thickness of the biological tissue to be bonded, tissue fusion can be optimized. Adjusting parameters like laser strength, frequency, and scanning speed can improve the tissue fusion effect [11].

Laser biologic tissue welding technology has been developed over decades and is trustworthy in terms of quick wound closure and mechanical strength. When laser power is too high or scanning time is too lengthy, severe thermal damage might occur [12], [13], [14]. Fried [15] found that when using an infrared laser to join full-length incisions in guinea pig dorsal skin with the addition of a biological dye to improve the mechanical properties, the tensile strength of laser-welded tissue incisions was high and increased with time of scanning, but both the epidermis and the dermis produced more severe thermal damage. The temperature of biological tissues rises when the laser acts on them. When the skin's temperature rises above its threshold, the unstable molecular crosslinks in the collagen of the skin tissue are gradually broken, resulting in macroscopically visible thermal contraction, a process known as thermal denaturation, and the tissue suffers various degrees of thermal damage [16], [17], [18]. As a result, minimizing heat damage to biological tissues while maintaining the mechanical properties of laser-welded biological tissues is a critical concern.

High-power continuous near-infrared laser connections have been proven to induce significant thermal damage to the aorta and skin tissue of living organisms [19], [20], [21], [22]. Continuous waves have been utilized to convey energy in previous skin attachment investigations, and energy transfer from the incision site to the surrounding tissue often leads to thermal injury to the *peri*-incisional tissue, which can cause a 500- μ m-wide area of damage. The optical breakdown is caused when the continuous laser power is sufficiently high above a threshold value [23], [24]. The high optical intensity required for optical breakdown can be achieved at the focal point of the femtosecond laser beam at high peak power, while laser pulses below this threshold level pass through the material without causing damage [25], [26], [27], [28], [29]. Femtosecond lasers have a narrow pulse width and high peak power. A continuous wave or long-pulsed laser's action on biological materials is based on linear absorption, but a femtosecond laser's action is nonlinear [30]. When the light intensity hits the threshold, a large amount of plasma is instantly generated and the absorption of photons increases dramatically [31]. As a result, in experiments, the average output power of

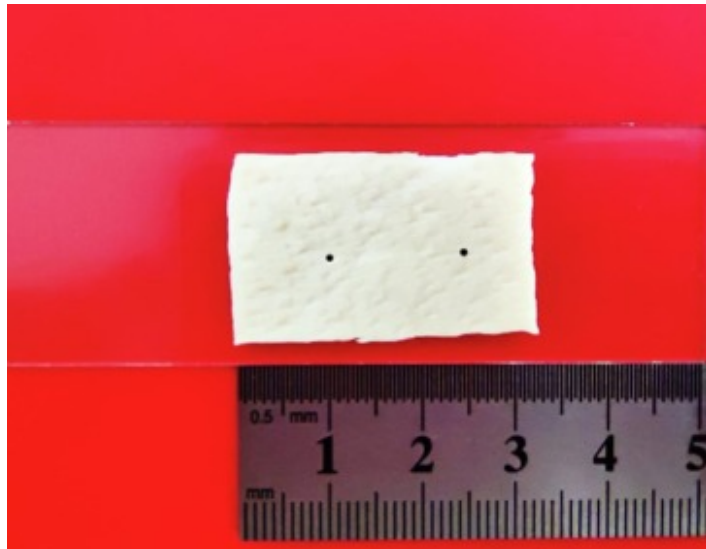
femtosecond lasers can be as low as a few tens of milliwatts, but the average output power of long pulses or continuous waves must be at least 1 W. For thermal destruction of biological tissues, high repetition rate femtosecond pulsed lasers require far lower temperatures than continuous lasers, which require 100°C. Song Ziyang et al. [32] used femtosecond laser to weld the incision of skin tissue in vitro to study the effect of femtosecond laser process parameters on the mechanical properties and thermal damage of skin tissue in vitro after welding. When biological tissue is welded with an ultra-short laser pulse, the water in the collagen helix absorbs the pulse energy and immediately rises to a much higher temperature than the surrounding tissue. This excitation breaks water-mediated interchain and intrachain hydrogen bonds and hydrophobic interactions and results in partial transient denaturation of collagen. As the water in and between the chains cools, hydrogen bonds and other bonds are re-established simultaneously in some reconstructed collagen helices. The use of ultrashort pulsed picosecond and femtosecond lasers to bond biological tissues can reduce thermal damage to surrounding tissues with a larger spot size [33]. Femtosecond lasers have many applications in biological imaging, cellular surgery, ablation of biological bodies, ophthalmic surgery, and dermatological treatment because of their precise action and low thermal damage when applied to biological tissues [34].

In this paper, we used a near-infrared femtosecond pulsed laser to weld in vitro skin tissue incisions, designed tests using an orthogonal test method, performed Masson staining on the bonded skin tissues, used an image texture feature method based on the grayscale co-occurrence matrix, extracted texture features from the skin microscopic tissues, and used the second-order statistical eigenvalues of the grayscale co-occurrence matrix as a criterion for evaluating the degree of thermal damage of femtosecond laser welding in vitro skin tissue.

2. Thermal damage process test

2.1. Test material

Based on the genomic similarity between pigs and humans, fresh isolated pig dorsal skin was selected for this experiment and made into a 30mm×20mm×2mm size specimen (including a subcutaneous fat layer of 0.5mm, as shown in Fig. 1). Soak the sample in a 5% bovine serum protein (BSA) solution for 15min before the test, then remove it and absorb the water on the sample's surface using gauze. Cut a 15mm long full-layer incision in the middle of the sample along 1/3 of the length direction, lay it on the slide, and keep the sample incision closed. The samples are laser-welded as soon as feasible to prevent moisture loss and diminished tissue activity.



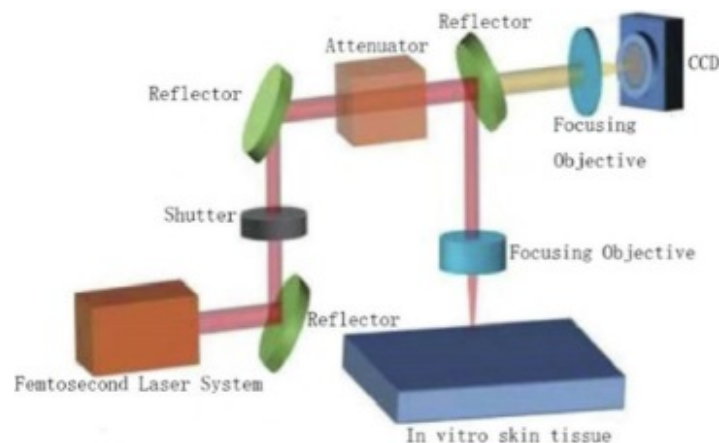
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Fig. 1. In vitro skin tissue samples.

2.2. Testing device

The femtosecond laser used in this study is the Monaco femtosecond laser manufactured by Coherent, which integrates a solid-state oscillator and amplifier at 1035nm wavelength to achieve a single pulse energy output of 40 uJ at a repetition frequency of 1 MHz with a maximum power of 40W. To meet the parameter requirements of different applications, Monaco can work flexibly from a single pulse output to full re-frequency output with a pulse width adjustable from 300fs to 10ps. The composition of the femtosecond laser is shown in [Fig. 2](#).

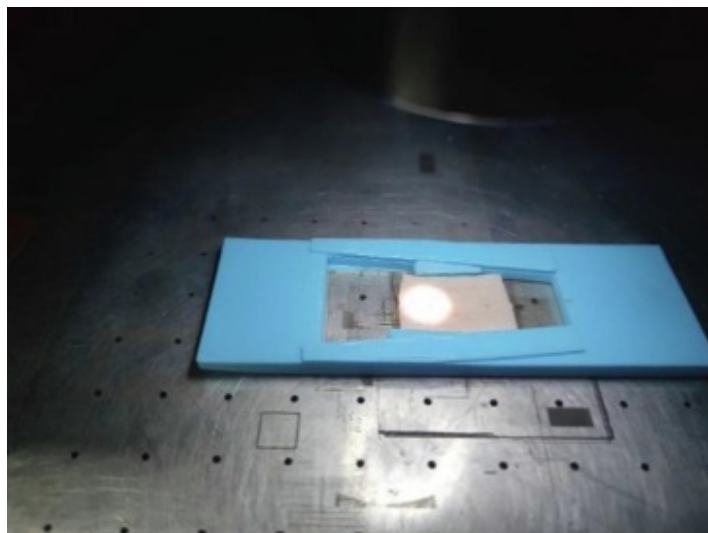


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Fig. 2. Schematic diagram of the femtosecond laser tissue welding device.

The processed samples are clamped using a fixture and placed on the laser carrier table. The clamped and placed in vitro skin tissue samples are shown in Fig. 3.



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Fig. 3. Clamped sample on the carrier table.

Tensile strength is a key criterion for determining the efficiency of laser-welded biological tissue fusion, and samples must be examined for it. As shown in [Fig. 4](#), an Epp Metrology F-30 digital display push-pull meter was used to evaluate the tensile strength of the sample slices. To satisfy the demand for tensile strength measurement in this study, the device is very sensitive, with an accuracy of 0.5 percent, and can automatically record the highest tensile force during the tensile process.



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Fig. 4. Tensile strength testing device.

An important indicator to evaluate the effectiveness of laser tissue welding is the thermal damage caused by the laser to the biological tissue during the fusion process. Since thermal damage is not directly accessible or measurable, but temperature and time affect the thermal damage of tissues, temperature measurements of biological tissue samples are needed. The temperature is monitored using a non-contact temperature measurement device, a thermal imager, as shown in [Fig. 5](#). By detecting the infrared radiation on the sample surface, and using signal processing and photoelectric conversion, the temperature distribution

on the sample surface is rendered into a visual image using different color systems. Thermal imaging cameras monitor the temperature of laser-welded in vitro skin tissue samples to avoid the effects of laser exposure and to monitor the temperature distribution in an area and the process of change. After the sample was placed on the work table, the infrared thermal imager was fixed at the shell of the laser. The distance of the lens from the sample was about 20cm. The focal length of the lens was adjusted to make the lens of the thermal imager focus on the skin tissue sample in vitro, and the temperature changes in the sample area were selected for analysis.



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Fig. 5. Temperature measurement equipment thermal imager.

The biopsy tissues were processed in sections after laser welding and the tissue sections were stained for observation using Masson trichrome staining method.

2.3. Experimental test method

There are numerous parameters affecting the laser connection of biological tissues. Laser power, laser wavelength, pulse duration, spot size, output mode, scanning mode, flux, heat source movement speed, and off-focus amount all have important effects on the degree of thermal damage and the fusion effect of biological tissues. In this study, four factors were selected as

independent variables: laser power(P), laser scanning speed(V), off-focus amount(L), and laser scanning times(N), with three levels designed for each factor, as indicated in Table 1. The laser power is the peak power of the femtosecond pulse laser, the laser scanning speed is the moving speed of the laser beam, the off-focus amount is the distance from the skin tissue surface of the focus, and the number of laser scanning is the number of times the laser beam completes the laser path. The test parameters were established using an orthogonal test to acquire the effective rule of a single factor on the fusion effect and thermal damage of in vitro skin tissue incision and to determine the combination of femtosecond laser parameters that make the optimum fusion effect of the incision. The test parameters are shown in Table 2.

Table 1. Table of parameter levels.

Test factors	Coding value	Coding Levels		
		1	2	3
P/W	X_1	12	14	16
V/mm·s ⁻¹	X_2	30	40	50
L/mm	X_3	-1	-0.5	0
N/freq	X_4	50	75	100

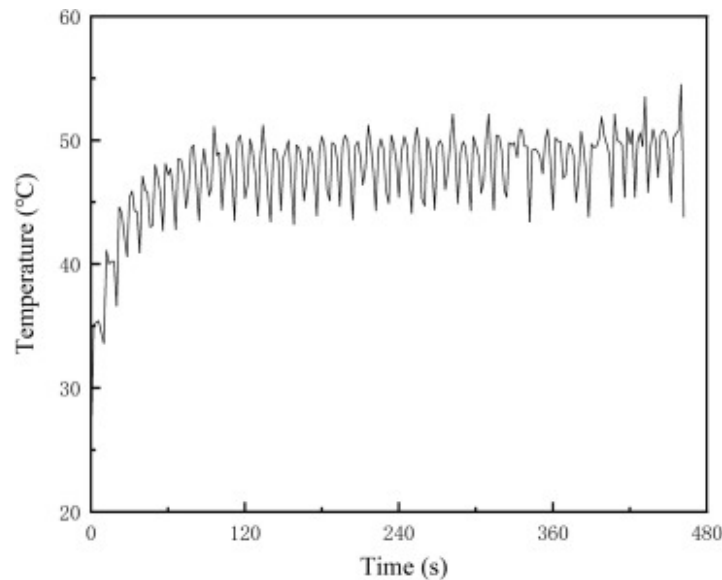
Table 2. Test Parameters.

No.	Laser power (W)	Laser scanning speed (mm/s)	Off-focus amount (mm)	Laser scanning times (freq)
1	12	30	-1	50
2	12	40	-0.5	75
3	12	50	0	100
4	14	30	-0.5	100
5	14	40	0	50

No.	Laser power (W)	Laser scanning speed (mm/s)	Off-focus amount (mm)	Laser scanning times (freq)
6	14	50	-1	75
7	16	30	0	75
8	16	40	-1	100
9	16	50	-0.5	50

After the bonded skin tissue was cooled, the samples were tested for tensile strength by clamping one side of the sample cut out with a jig, and the other side was clamped evenly using two glass pieces of the same width as the sample cut out so that the sample cutout was evenly stressed during the stretching process. After pushing the pull-push meter to zero, the glass piece was slowly pulled in the direction perpendicular to the incision until the sample incision was completely pulled, and the maximum tensile strength of the in vitro skin tissue sample after laser welding was calculated by recording the maximum tensile value displayed on the pull-push meter.

The peak temperature profile during laser tissue welding was obtained by the thermal imager, as shown in [Fig. 6](#).



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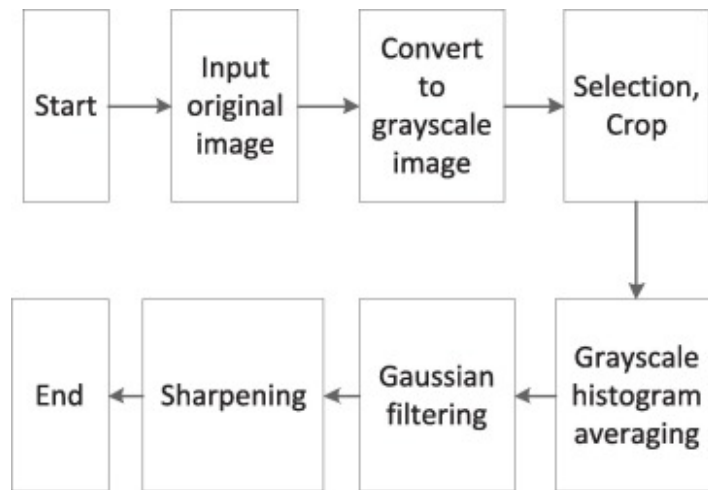
Fig. 6. Peak temperature-time curve.

As can be seen from Fig. 6, the peak temperatures of biological tissues measured using a thermal imager in this study ranged from 40 to 110°C. The main operating temperature of laser tissue welding was 60°C for protein and collagen denaturation effects, and 80°C and 100°C respectively for tissue chemical concentration equilibrium failure temperature and tissue water molecule vaporization temperature. The thermal damage of the laser on biological tissue is related to the temperature of the tissue during the action and the duration at that temperature; the higher the peak temperature of the skin tissue, the greater the degree of thermal damage. During the connecting process, the temperature change on the surface of the in vitro skin is measured using a thermal imager, which cannot reliably assess the thermal damage in the dermis beneath the epidermis. Therefore, the texture feature extraction method based on the grayscale co-occurrence matrix was used to extract texture features from the in vitro skin microstructure under different laser scanning conditions, and the second-order statistical eigenvalues of the grayscale co-occurrence matrix were used as an index for evaluating the degree of thermal damage of the in vitro skin subjected to laser scanning and to analyze the changing pattern of biological microstructure under the laser.

The grayscale co-occurrence matrix (GLCM) is defined as the probability of leaving a pixel point with grayscale value i' on a point at a fixed location (separated by distance d and offset angle θ) from a pixel point with grayscale i . The values of all probabilities can be expressed in the form of a matrix, which is called the grayscale co-occurrence matrix. Gray-level co-occurrence matrix (GLCM) was very popular in the extraction of texture features, and the features extracted by this method had the excellent discriminating ability. The GLCM method proposed by Haralick et al. [35] has become a classic algorithm for texture feature extraction and has been widely used in many fields.

Haralick [36] has proposed 14 statistics calculated based on GLCM: angular second-order moments, entropy, contrast, homogeneity, correlation, variance, sum average, sum variance, sum entropy, difference variance, difference average, difference entropy, correlation information measure, and maximum correlation coefficient. Among the 14 texture features, 4 features are easy to compute and work well, namely, angular second-order moments, entropy, correlation, and contrast. The angular second-order moments are the sum of squares of the elements of the grayscale co-occurrence matrix, reflecting the uniformity of the grayscale distribution and texture coarseness. Entropy measures the randomness of the image texture. Correlation measures the similarity of the elements of the spatial grayscale matrix in the row or column direction, and the magnitude of the correlation value reflects the local grayscale correlation in the image. Contrast is the contrast near the main diagonal of the grayscale co-occurrence matrix, which reflects the sharpness of the image and the depth of the texture grooves.

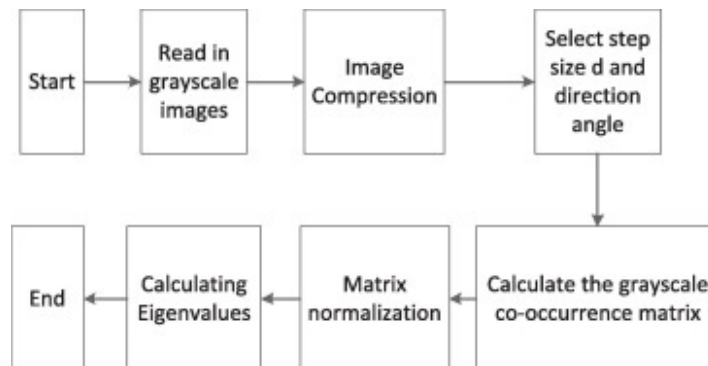
The method requires the use of Image-J software to pre-process the images and eliminate the negative impacts of factors such as background brightness and ambient light on the test results, as depicted in Fig. 7. To extract texture features from microscopic images, the related code was built in the MATLAB program, and the algorithm flowchart is given in Fig. 8.



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Fig. 7. Image pre-processing flow chart.



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Fig. 8. Flow chart of grayscale co-generation matrix algorithm.

3. Results and discussion

3.1. Analysis of tensile strength

Tensile tests were performed on nine skin samples using a push-pull meter, and the ratio of the maximum tension to the cross-sectional surface area of the incision was recorded as the tensile strength of the incision, and the cross-sectional surface area of the incision was the product of the length of the incision and the thickness of the skin tissue. The tensile strength values obtained from in vitro skin tissue incisions after welding are shown in Table 3. The extreme differences between the four process parameters and the tensile strength were calculated as shown in Table 4.

Table 3. Tensile strength values obtained from process tests.

No.	Laser power (W)	Laser scanning speed (mm/s)	Off-focus amount (mm)	Laser scanning times (freq)	Tensile strength (N/cm ²)
1	12	30	-1	50	0.67
2	12	40	-0.5	75	1.9
3	12	50	0	100	1
4	14	30	-0.5	100	2.6
5	14	40	0	50	1.47
6	14	50	-1	75	1.2
7	16	30	0	75	9.67
8	16	40	-1	100	5.23
9	16	50	-0.5	50	6

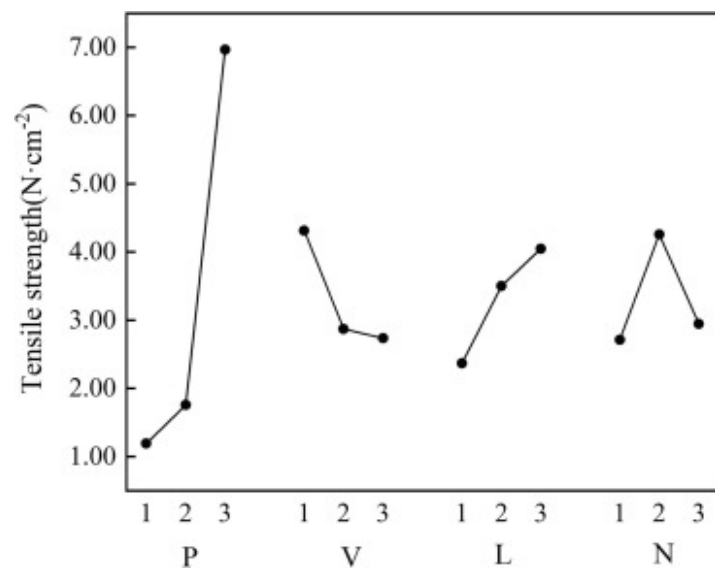
Table 4. Four-factor extreme difference analysis of tensile strength.

	Laser power	Laser scanning speed	Off-focus amount	Laser scanning times
K ₁	3.57	12.94	7.1	8.14

	Laser power	Laser scanning speed	Off-focus amount	Laser scanning times
K₂	5.27	8.6	10.5	12.77
K₃	20.9	8.2	12.14	8.83
k₁	1.19	4.31	2.37	2.71
k₂	1.76	2.87	3.50	4.26
k₃	6.97	2.73	4.05	2.94
R	5.78	1.58	1.68	1.54

As shown in Table 3, the difference between the tensile strength of the in vitro skin tissue incision at a laser power of 16W and the tensile strength at lower power is significant, and the skin incision fusion with better connection performance can be obtained at higher power. The tensile strength of the in vitro skin tissue sample incision can achieve 9.67N/cm² using the parameters laser power of 16W, scanning speed of 30mm/s, off-focus amount of 0, laser scanning 75 times, the skin incision is well fused, and the mechanical characteristics are high. The laser power was much greater than the remaining three factors, indicating that the laser power was the main factor influencing the mechanical properties of the fusion of femtosecond laser-welded skin tissue incisions. The heat transported to the inside of the in vitro skin tissue increases when the laser power is increased, and the water molecules within the tissue gain energy, causing more collagen triple helixes to dissolve and reconstitute, increasing tensile strength.

The graph of the extreme difference analysis of the four laser process parameters on the tensile strength of skin tissue is shown in Fig. 9. Since the B, C, and D factors at the same level of the A factor are in the same place but are paired differently in the orthogonal test, the influence of different levels of the A factor, for example, can be roughly canceled out because the orthogonal test is orthogonal and integrally similar. In this way, we can consider the change in $\bar{T} - T_{Ai}$: (the mean of the data at level i of factor A) as being caused by different levels of factor A. R_A can then be approximated as the extent to which a change in the level of factor A can cause a change in the whole experiment. The higher the value of R_A , the greater the influence of the A factor on the whole experiment, and the more important the factor is to the experiment, and vice versa. As can be seen from Table 4 and Fig. 9, the laser power has the most significant effect on the tensile strength of the skin tissue incision after attachment.

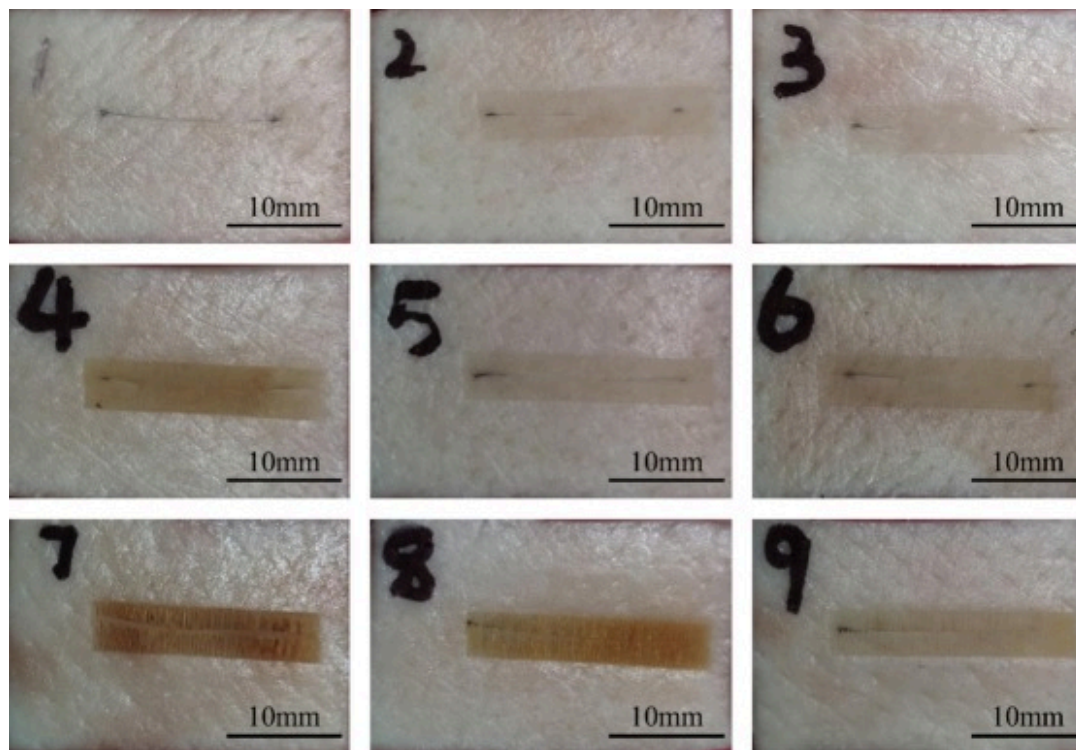


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Fig. 9. Trend graph of four factors and skin tissue incision tensile strength values.

A photograph of the in vitro skin tissue incision after the connection of nine sets of laser parameters is shown in [Fig. 10](#).



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Fig. 10. Appearance of the in vitro skin incision after connection.

To observe the appearance of the skin tissue incision in Fig. 10, the laser power has a large effect on the degree of water loss in the skin tissue; at a laser power of 12W, there was slight water loss at the incision, but no serious scarring was seen, and the epidermal layer produced a better connection. The water loss was exacerbated when the power was increased to 14W, and a light brown scar with a width of about 3mm was seen around the incision, with both the epidermis and the dermis connected, with a tensile strength of up to 1.5N/cm^2 . When the power was set to 16W, the skin tissue incision lost water severely and a scorched yellow scar was seen at the incision, and both epidermis and dermis produced better fusion with tensile strength up to 9.67N/cm^2 . When the skin is subjected to a slow laser scan, the absorption and scattering effect of the laser on the tissue as the spot moves tends to overlap with the laser action in the area irradiated by the previous spot, resulting in increased coagulation

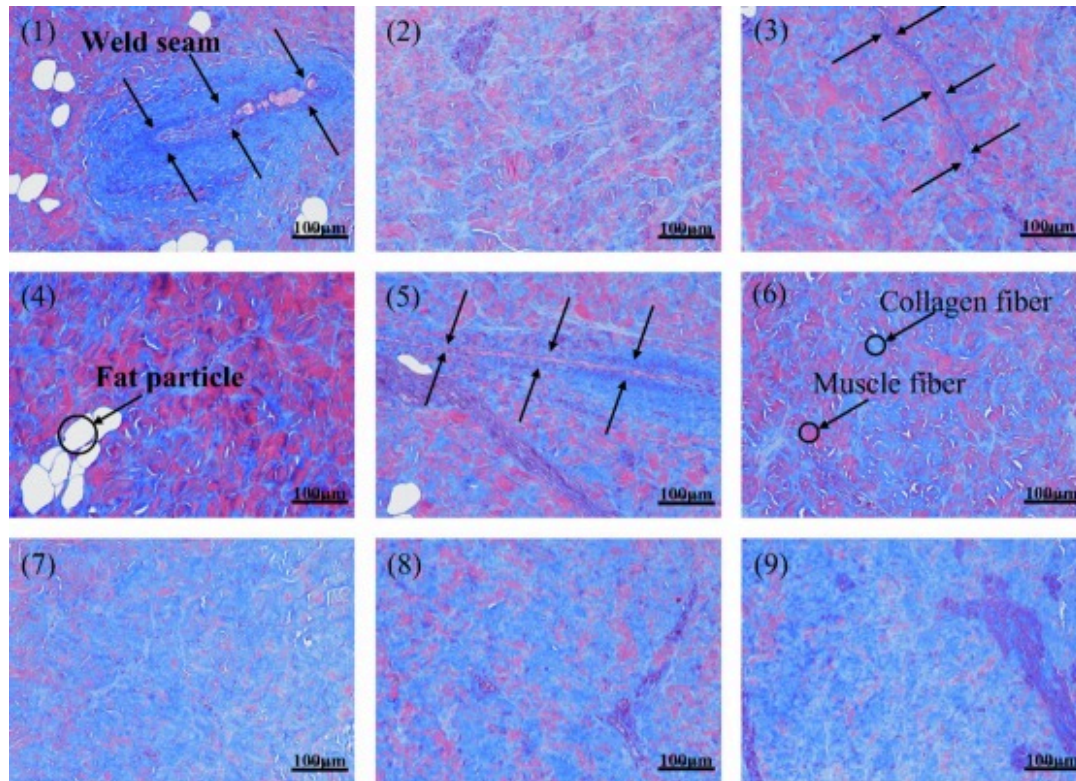
and fusion of the tissue. As shown in (4)(5)(6) in Fig. 10, the smaller the scanning speed, the greater the degree of water loss from the incision surface and the more pronounced the scar.

In the laser welding process, the power density of the spot center at the laser focal point is too high, which tends to make the sample evaporate into a hole, and the power density distribution is relatively uniform in all planes leaving the laser focal point. As shown in Fig. 10(3), when the off-focus amount is 0mm, the laser focus is on the surface of the in vitro skin tissue, and the laser energy is gathered at the incision on the upper surface, which makes the incision rapidly-produce water loss and cracking, and as the off-focus amount increases, the laser focus leaves the upper surface of the sample and a good connection is formed at the incision.

As the number of laser scans increases, the heat accumulation effect is enhanced, more collagen within the tissue acquires remodeling and the tensile strength increases. When the laser scanning times increase to 100, biological tissues are exposed to air for a long time, water molecules inside biological tissues decompose to participate in collagen remodeling sessions, and biological tissues are highly susceptible to dehydration and carbonization, making the tensile strength of skin tissue incisions lower.

3.2. Microstructure analysis

The middle of the in vitro skin tissue incision after welding was taken and made into sections of 2–3 μ m thickness by the paraffin sectioning method. The sections were stained by Masson trichrome staining method, and the microscopic tissue of the incision after the connection of 9 groups of parameters is shown in Fig. 11.



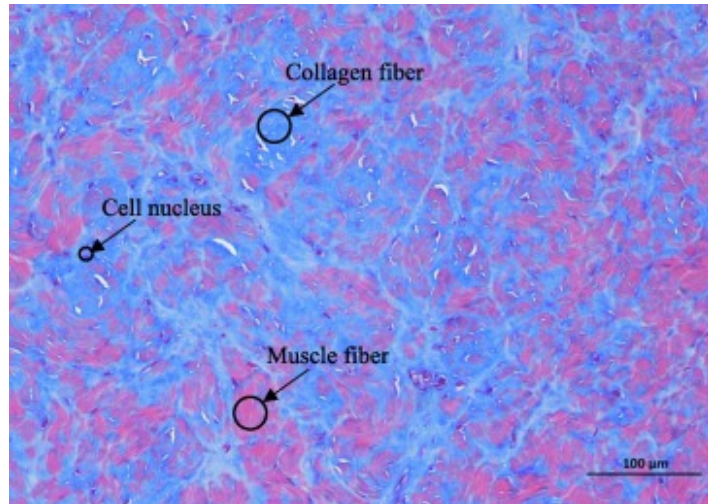
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Fig. 11. Microscopic tissue section of the incision after welding.

Microscopic tissues showed blue or green color for collagen fibers and red color for muscle fibers after Masson staining. The major components involved in the welding process in skin tissue are water, collagen, and elastin. During laser welding, water molecules in the tissues heat the collagen helix by absorbing laser energy. When the temperature of collagen tissue increases, hydrogen bonds, hydrophobic interactions, and electrostatic interactions in biological tissues are disrupted, the collagen triple helix undergoes partial dissociation and the collagen molecule undergoes reversible denaturation. The protein molecules within the living tissue recombine to create collagen helices as the tissue cools.

The microstructure of the in vitro skin without laser scanning is shown in Fig. 12, and the percentage of collagen fibers in the skin tissue without laser scanning was calculated as 48.47% by separating the red and blue parts of the microstructure image using Image-J software. The collagen fibers of the skin tissue are evenly distributed, with blue collagen fibers wrapping around red muscle fibers and cell clusters, with obvious color contrast, clear texture, and directional fiber arrangement.



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Fig. 12. Microstructure of in vitro skin without laser scanning.

As shown in Fig. 11(1), the broken edge of the incision at the connection is visible, and unlike the in vitro skin tissue without laser scanning, the red muscle fibers tissue is decreasing, and the collagen fiber structure is changing, showing a sparse and dense state, and the skin tissue texture is more blurred near the incision connection.

It is not easy to distinguish the incisional junction in Fig. 11(2), the collagen fiber distribution becomes denser, the color contrast is not obvious, the tissue texture is blurred, and the collagen fiber texture distribution has directionality.

Fig. 11(3) shows that collagen deposition was partially formed at the incision junction, and the collagen fiber arrangement near the incision did not change significantly, with obvious color contrast and uniform collagen fiber distribution and directional arrangement.

In Fig. 11(4), the muscle fibers accounted for a larger proportion of the incisional junction, and the majority of the incisional junction was better fused, with aggregation of muscle fibers near the incision, reduced collagen fibers, and disorganized fiber distribution without directionality.

In Fig. 11(5), there is obvious collagen deposition at the incision, the collagen fibers are aggregated near the incision with higher density, the muscle fibers are distributed outside of the collagen fibers, the color contrast is obvious, and the collagen fibers and muscle fibers are distributed with directionality.

A better fusion was formed at the incision in Fig. 11(6), and the location of the incision was not easily distinguished under the microscope, with a uniform and orderly distribution of collagen fibers and muscle fibers with directionality, and obvious color contrast, and clear microscopic tissue texture.

In Fig. 11(7), there is a large amount of collagen formation deposition, a decrease in the content of muscle fibers, a chaotic fiber distribution, a microstructure with blurred texture, and weaker color contrast.

The incisional junction shown in Fig. 11(8) is not easily distinguishable, producing a small amount of collagen deposition with a high density of collagen fibers and a high degree of disorganization of fiber distribution without directionality.

Fig. 11(9) shows that collagen deposition is produced near the incision, muscle fiber content is reduced, fiber distribution is somehow directional, the color contrast between collagen fibers and muscle fibers is weak, and the microstructure texture is blurred.

3.3. Analysis of thermal damage

The temperature change at the incision region was tracked using a thermal imager during the laser tissue welding process using the 4th set of parameters, and the temperature-time curve was obtained as shown in Fig. 6. When the temperature curves of the femtosecond laser and the continuous laser connection to the in vitro skin tissue are compared, it can be seen that the femtosecond laser produces far less thermal damage than the continuous laser. When a femtosecond laser with a high repetition frequency is applied to biological tissue, free electron-mediated chemical effects, photochemical bond breakage, and thermal effects are produced in the tissue. Because the thermal diffusion distance does not reach the light penetration depth when the laser pulse width is smaller than the thermal relaxation time, thermal damage to surrounding tissues can be avoided. As can be seen from Fig. 6, the tissue temperature during femtosecond laser tissue welding is approximately 40–100°C. According to the

rule of variation of the composition in the tissue, the thermal effect produced by the tissue in this temperature interval can be attributed to the change of molecular composition, which is consistent with the mechanism of action of femtosecond laser welding of biological tissues.

The calculated microstructure texture characteristic values were analyzed using the orthogonal test method and Minitab software to derive the effects of femtosecond laser parameters on tissue texture and thermal damage.

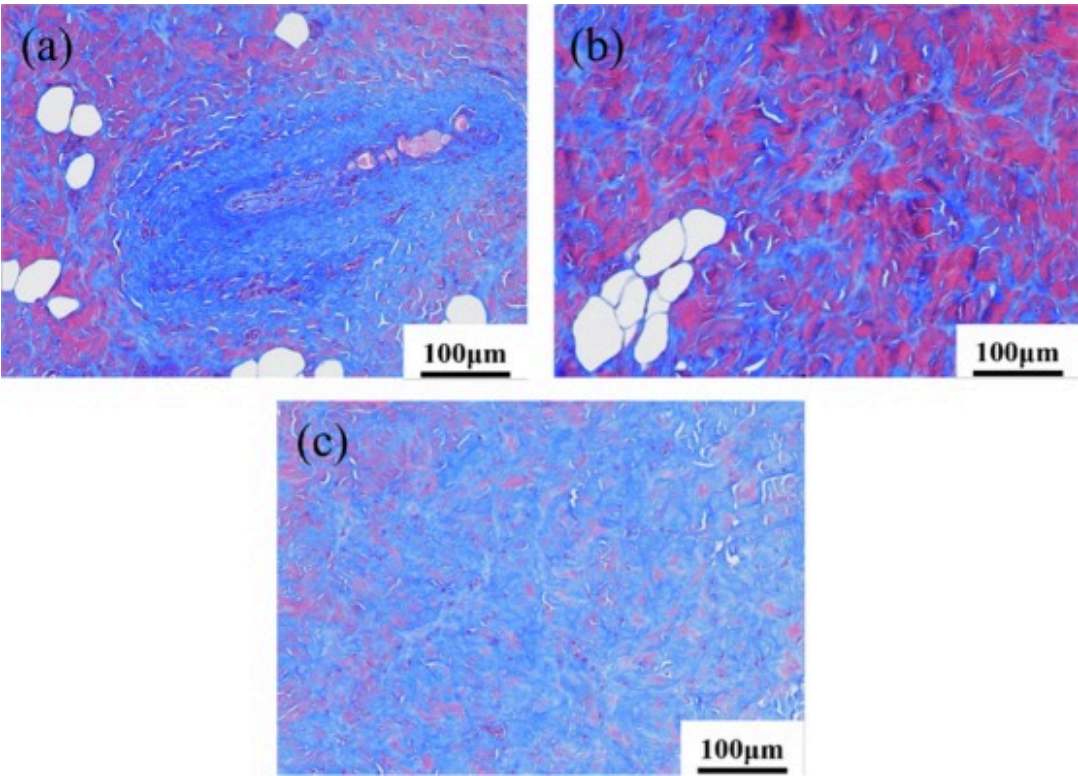
From [Table 5](#) and the peak temperature extreme difference analysis, it can be concluded that the laser power has the most significant effect on the peak temperature of the in vitro skin tissue surface. When the laser power is 12W and 14W, the peak temperature of the skin sample surface is in the range of 40–80°C. In this temperature range, the protein molecules and collagen molecules in the skin tissue are gradually unfolded, and with the continuous effect of the laser, the protein and collagen in the contact part of the incision are newly entangled and overlapped, and the incision of the skin tissue is connected. When the laser power is 16W, the peak temperature distribution of the skin sample surface is 50–110 °C, and the protein denaturation of the skin tissue occurs near 60 °C. Tissue coagulation occurs when the temperature exceeds 60°C, and the tissue cells scanned by the laser will also be necrotic, the femtosecond laser has more thermal damage to the isolated skin tissue. The effect of scanning speed on the in vitro skin tissue is secondary to the laser power, the peak surface temperature of the isolated skin tissue decreases as the scanning speed increases as shown in [Table 5](#). The superimposed effect of the laser in the adjacent scanning area was more significant at low scanning speed, with an increase in tissue surface temperature.

Table 5. Calculation results of characteristic parameters.

No.	P(W)	V(mm/s)	L(mm)	N(freq)	T(°C)	Angular second-order moment	Entropy	Contrast	Relevance
1	12	30	−1	50	50.3	0.0064	5.2534	19.5693	0.0202
2	12	40	−0.5	75	46.9	0.0070	5.1907	17.9286	0.0217
3	12	50	0	100	45.7	0.0075	5.1496	17.7692	0.0257
4	14	30	−0.5	100	78.5	0.0067	5.1932	17.5699	0.0269
5	14	40	0	50	50.1	0.0075	5.1306	16.2762	0.0243
6	14	50	−1	75	49.9	0.0063	5.2729	22.4873	0.0119

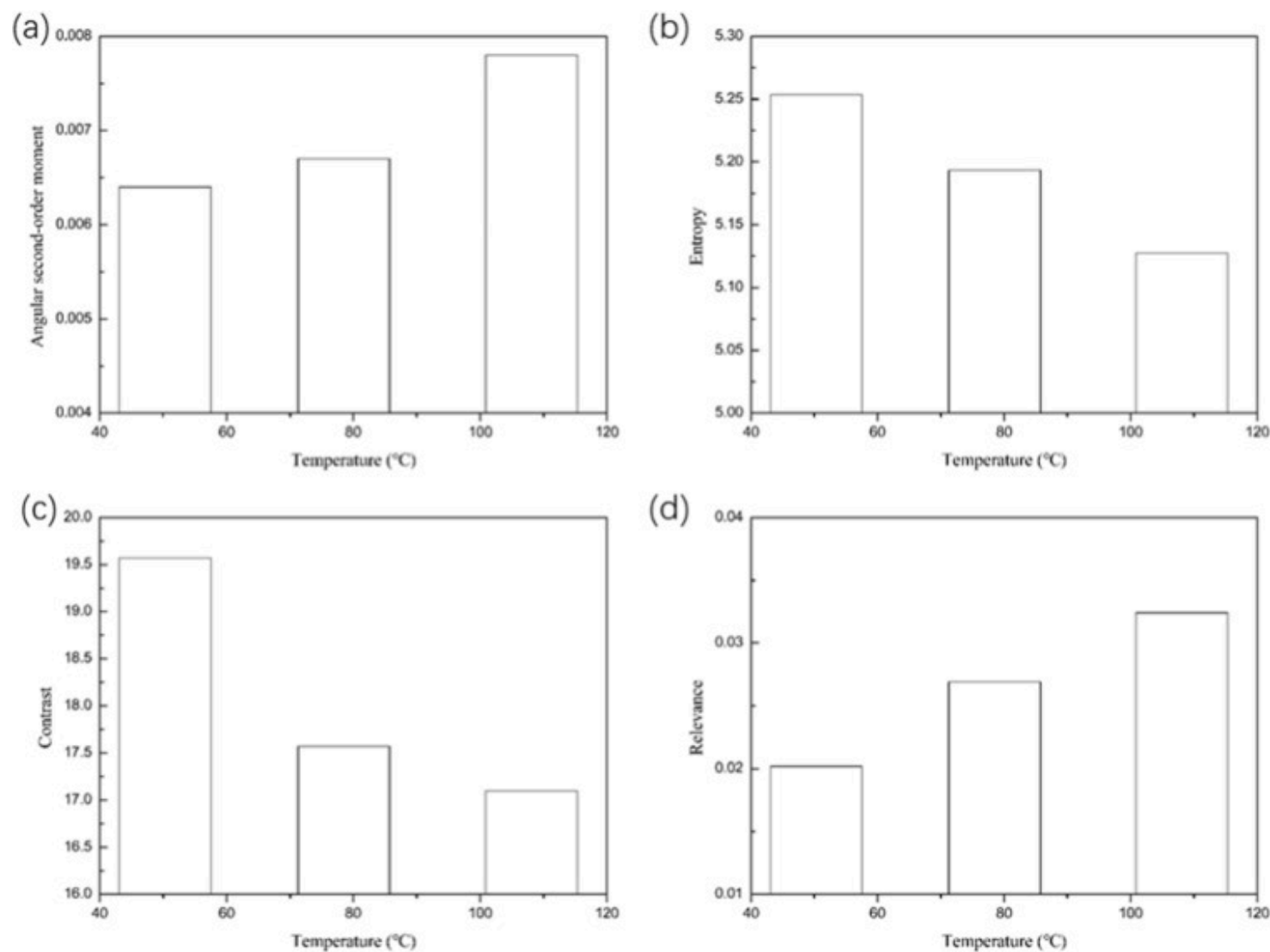
No.	P(W)	V(mm/s)	L(mm)	N(freq)	T(°C)	Angular second-order moment	Entropy	Contrast	Relevance
7	16	30	0	75	108.1	0.0078	5.1274	17.0993	0.0324
8	16	40	-1	100	82	0.0070	5.1636	15.0136	0.0311
9	16	50	-0.5	50	54.5	0.0065	5.2124	15.5515	0.0307

The peak temperatures of groups 1, 4, and 7 were selected, corresponding to the temperature at which the protein molecular structure changes, protein denaturation condensation, and vaporization during laser tissue welding. The microstructure images are shown in Fig. 13, and the texture eigenvalues were extracted to obtain the change map in Fig. 14.



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Fig. 13. Microstructure of skin tissue incision bonded with process parameters in group 1, group 4, and group 7.



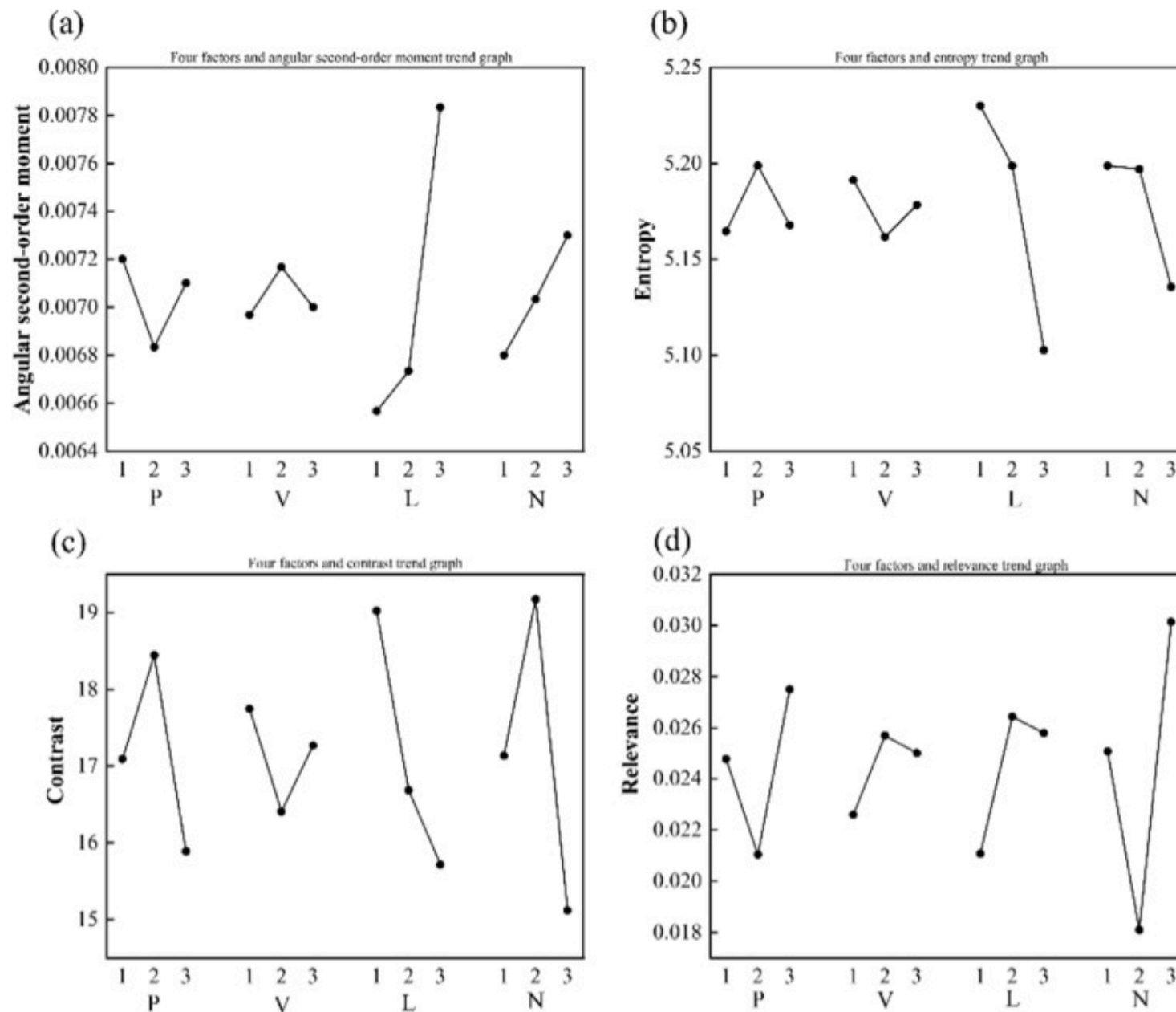
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Fig. 14. Variation of microstructure characteristic values after welding with different process parameters(a: Angular second order moment, b: Entropy, c: Contrast, d: Relevance).

As can be seen in Fig. 14, the angular second-order moment and correlation of the grayscale co-occurrence matrix eigenvalues increase with increasing temperature during laser tissue welding, while the entropy and contrast decrease with increasing temperature. According to the surface of the textural features characterized by the eigenvalues, the tissue texture is more homogeneous and uniform when the surface temperature is lower, with deep grooves and high clarity of texture display. As the temperature increases, the texture of the tissue becomes disordered, the arrangement more complex and the grooves of the texture increasingly shallow. As shown in Fig. 13, it can be found that the boundary between the tissue textures gradually becomes blurred (as shown in Fig. 13(c)) from the fine-grained clarity in Fig. 13(a). This is consistent with the pattern of changes that occur as the temperature increases skin tissue begins to undergo partial tissue superheat necrosis, with denaturation and coagulation occurring near 60°C. Since the thermal imager can only capture the temperature of the sample surface, the study of thermal damage to the dermis of skin tissue can be derived from the variation pattern of microstructure characteristic values with temperature.

Fig. 15 shows a plot of the effect of the parameters on the microstructure feature values of the in vitro skin tissue samples bonded by the femtosecond laser. From the trend plots and the extreme difference analysis, it can be concluded that the off-focus amount and the laser scanning times have a significant effect on the microstructure feature values, which means that these two factors have a greater influence on the thermal damage inside the laser-bonded skin tissue.



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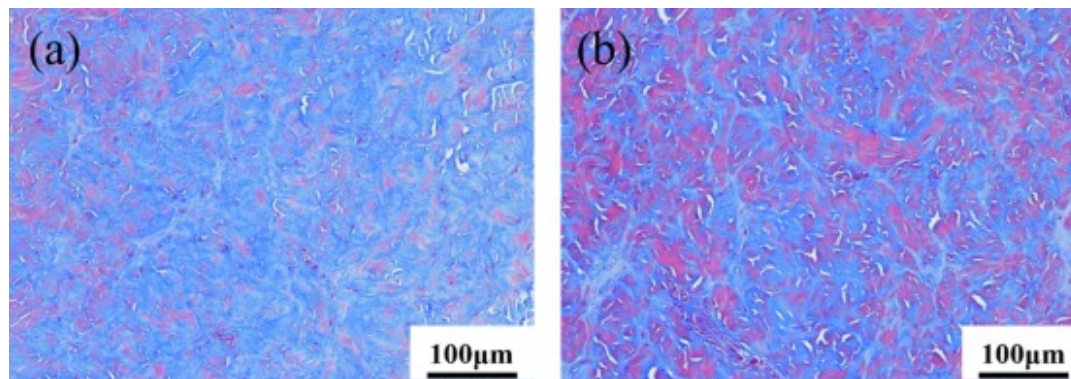
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Fig. 15. Laser parameters and microstructure feature values effect diagram.

When the off-focus amount is 0, the laser focus stays on the surface of the skin tissue, and the thermal relaxation time of the biological tissue is about $1\mu\text{s}$. The femtosecond laser pulse width is less than $1\mu\text{s}$, the heat is too late to be transmitted to the dermis, and the damage stays in the epidermis. By increasing the amount of off-focus, the laser focus is in the dermis and heat can be transmitted to the dermis, which increases the amount of thermal damage within the skin tissue.

The absorption of the femtosecond laser on biological tissue is nonlinear. The temperature of biological tissue rises rapidly as processing time increases, and if the duration at high temperature is too long, the tissue will be necrotic and the water molecules will be vaporized, which will easily cause mechanical rupture and thermal decomposition of the tissue fragments and eventually make the tissue carbonized.

When the laser power was 16W, the scanning speed was 30mm/s, the off-focus amount was 0, and the laser was scanned 75 times, the peak temperature during the laser welding was as high as 108.1°C . As shown in Fig. 10(7), a burnt-yellow scar was produced at the skin tissue incision, and water loss deformation was produced at both edges of the incision. The angular second-order momenta and correlations of the microstructure texture are larger, and the entropy and contrast values are smaller. As shown in Fig. 16(a), the microstructure has uneven grayscale distribution, coarse texture, shallow grooves, non-uniform distribution of texture, and non-directional distribution of collagen fibers, indicating the normal organization of the microstructure is disrupted. Although the tensile strength of the bonded incision is greater at this time, vaporization of water molecules inside the tissue occurs and thermal damage to the tissue is greater.



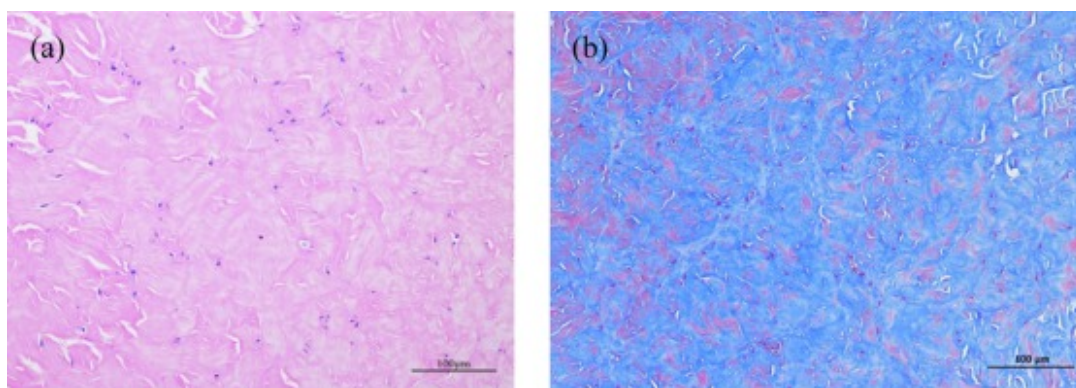
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Fig. 16. (a) High thermal damage value microstructure (b) Low thermal damage value microstructure.

When the laser power was 14W, the scanning speed was 50mm/s, the off-focus amount was -1mm and the laser was scanned 75 times, as shown in Fig. 10(6), the skin tissue incision produced a better connection, the epidermis was slightly discolored, no scar tissue was produced, and no water loss bulges were seen. The angular second-order momenta and correlation of the microstructure texture were smaller, and the entropy and contrast were larger. As shown in Fig. 16(b), the microstructure texture was clear, the color contrast was obvious, the collagen fibers had certain directionality in distribution, no collagen deposition was produced, and the microstructure was similar to the structure of the skin tissue that was not scanned by the laser, and the thermal damage inside the tissue was smaller at this time. The internal thermal damage pattern of the tissue exhibited by the microstructure feature values is consistent with the pattern presented by the peak temperature.

A continuous laser with a power of 4W and a scanning speed of 75mm/s was used to connect the in vitro skin tissue incision with a processing time of 20min, and the tensile strength of the connected incision was 9.95N/cm². The tensile strength is similar to that obtained from the test parameters of group 7 in this study. The images of the microstructure of the incisions obtained from these two groups of tests are shown in Fig. 17(a) (b), and Table 6 shows the image texture feature values of the two groups of microstructures.



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Fig. 17. (a) Continuous laser-bonded in vitro skin microstructure (b) Femtosecond laser-bonded in vitro skin microstructure.

Table 6. Microstructure texture feature values of in vitro skin with different laser connections.

	Tensile strength(N/cm ²)	Angular second-order moment	Entropy	Contrast	Relevance
Continuous Laser	9.95	0.0129	4.68	11.6	0.022
Femtosecond Laser	9.67	0.0078	5.1274	17.0993	0.0207

From Fig. 17 and Table 6, it can be seen that the microstructure texture distribution of the continuous laser-welded tissue is not uniform, the distribution of the texture has randomness, the directionality is weak, and the grooves of the texture are deep. When the continuous laser and the femtosecond laser produce the same mechanical properties of the incision fusion when connecting the in vitro skin tissue, the femtosecond laser produces less thermal damage.

Laser tissue welding with the continuous laser takes advantage of the thermal mechanism of laser interaction with biological tissue to partially denature and recover collagen. During the welding process, water in the tissue absorbs the laser energy and heats the collagen helix. When the temperature of collagen tissue rises above 60°C, the hydrogen bond, hydrophobic interaction, and electrostatic interaction are destroyed, and part of the collagen triple helix will temporarily dissociate, and then re-form the same bond of biological histamine molecules as the tissue cools. The use of femtosecond lasers to weld biological tissues can induce certain biochemical and physiological reactions at the molecular level in biological systems that induce tissue binding and affect the wound healing process and tissue structural reorganization. Both femtosecond and continuous lasers can alter tissue structure and cause damage through linear and nonlinear absorption, but femtosecond lasers bond biological tissues with a deeper penetration depth than continuous lasers. Femtosecond laser reduces the thermal damage to living cells when the laser bonds biological tissues due to its non-thermal mechanism of molecular cross-linking, thus reducing the postoperative inflammatory response.

4. Conclusions

The femtosecond laser can deliver energy to the inside of biological tissues, heating collagen molecules to cause reversible denaturation and partial deconstruction. The collagen spiral reorganizes as the tissues cool, allowing for rapid connection closure of in vitro skin tissues with the intensity required for trauma closure.

Laser power significantly affects the tensile strength of the bonded incision, with an increase in power followed by an increase in tensile strength. Quantifying thermal damage in skin tissue using eigenvalues from a grey-scale co-occurrence matrix. The thermal degradation of laser-bonded skin tissue is more affected by the amount of off-focus and the laser scanning times. The off-focus amount increases, the depth that can be reached by heat transfer increases, and the thermal damage to skin tissue increases. The laser scanning times increases, the heat accumulation increases, and the thermal damage generated within the tissue increases.

The correspondence between peak temperature and microstructure characteristic values were established by studying the fusion morphology, mechanical properties, and thermal damage of femtosecond laser-bonded in vitro skin incisions. The results showed that the better effect of in vitro skin tissue incision welding and the smallest thermal damage value were achieved at a laser power of 14W, a scanning speed of 50mm/s, an off-focus amount of -1 mm, and 75 laser scanning times.

5. Author statement

I have made substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work; AND

I have drafted the work or revised it critically for important intellectual content; AND

I have approved the final version to be published; AND

I agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

All persons who have made substantial contributions to the work reported in the manuscript, including those the provided editing and writing assistance but who are not authors, are named in the Acknowledgements section of the manuscript and have given their written permission to be named. If the manuscript does not include Acknowledgements, it is because the authors have not received substantial contributions from non-authors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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